



FPSO ALBACORA - BOT
PROJECT No. 20250501

PIPING MATERIAL SPECIFICATION

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1. INTRODUCTION

1.1. PROJECT BACKGROUND

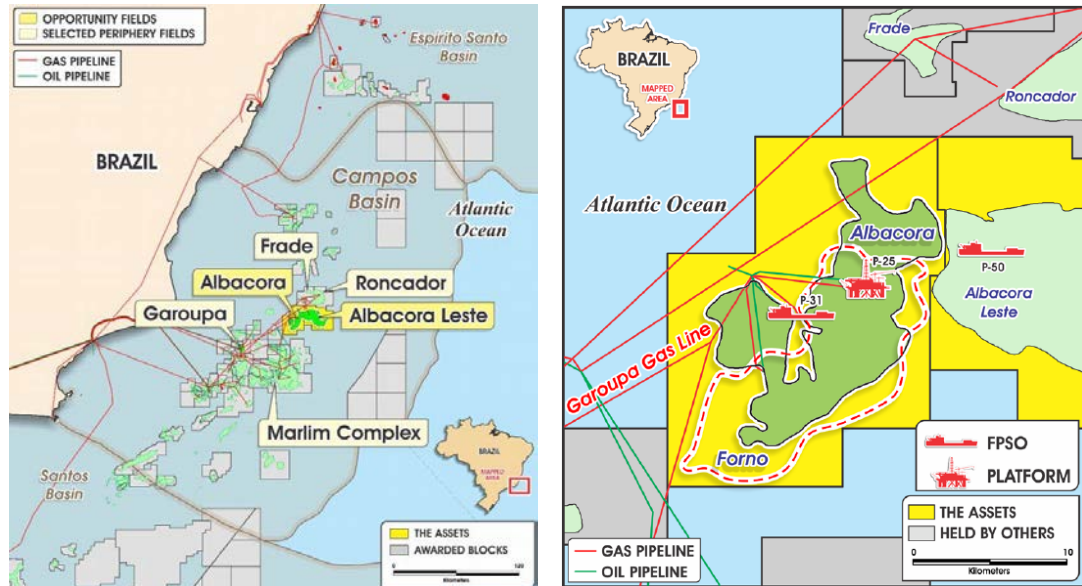


Figure 1-1: Albacora Field

Albacora is a brownfield project. Petrobras discovered Albacora in September 1984 and started output from the field in October 1987.

The Albacora field has an area of 455 km² and is located in the northern area of the Campos Basin, in water depths ranging from 100 to 1,050 m, at a distance of about 110 km from Cabo de São Tomé, on the northern coast of the state of Rio de Janeiro. Petrobras is the operator of the field with 100% participation.

High quality 27° API oil from the high porosity, high permeability Albian Namorado Sandstones and Oligocene and Miocene Carapebus turbiditic reservoirs, and ~30° API oil for the pre-salt reservoir.

It comprises 65 wells, two production units, and seven manifolds – six for oil production and one for water injection.

Petrobras has two units in operation at Albacora — the P-25 semi-submersible production platform and the P-31 FPSO.

The semi-submersible platform P-25 has a process plant with a capacity of 100,000 bopd and 115 mmcf/d of natural gas. There are 14 production wells connected directly to the platform and 16 wells connected to two production manifolds by a vertical flowline connection system.

The FPSO P-31 process plant capacity is 200,000 bopd and 130 mmcf/d of gas. The plant processes the oil produced in the P-25 platform and the oil production of 28 production wells connected to four production manifolds.

Oil is offloaded through shuttle tankers and gas is exported through the PGP-1 platform at the neighboring Garoupa field.

As part of the revitalization project, of the field, the two units are expected to be decommissioned from the field to make room for the new FPSO.

The new Albacora FPSO shall consider 25 wells (including 15 oil producers and 10 water injectors) in the design, moored offshore Brazil at a location with water depth of 670 meters and designed for an uninterrupted operation (without the need for drydocking) design life of at least 20 years.

The proposed FPSO has a total liquid handling capacity of about 31,850 Sm³/d max. The oil processing capacity, the total gas handling capacity and the water injection capacity are around 19,079 Sm³/d, 6,000,000 Sm³/d and 22,000 Sm³/d respectively.

1.2. OBJECTIVE

This specification stipulates the minimum requirements for the Design, Selection of materials, fabrication, manufacture & supply of piping components & valves for use in FPSO-TOPSIDES.

The requirements set forth do not relieve vendor of his responsibility to perform all services in safe manner, to meet applicable codes and standards, to comply with government regulations, to meet class requirements and to supply product/services capable of performing its intended service.

1.3. DEFINITIONS

The following words and expressions shall have the meanings hereby assigned to them except where the context otherwise requires.

Term	Meaning
BUYER	Petróleo Brasileiro S.A (Petrobras)
SELLER	Shapoorji Pallonji Energy Private Limited (SPE)
SUB-CONTRACTOR / VENDOR	Contract placed by SELLER for any portion of the CONTRACT or work sublet (to Subcontractor or Vendor)
CLASS	Classification Society for the FPSO
Shall	This word is to be understood as a mandatory requirement.
Should	The word is to be understood as not mandatory but is recommended.
May	Used when alternatives are equally acceptable

1.4. ABBREVIATIONS

Abbreviation	Definition
SPOG	Shapoorji Pallonji Oil and Gas Private Limited
ABS	American Bureau Shipping
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
CA	Corrosion Allowance
CS	Carbon Steel
CuNi	Copper Nickel
DSS	Duplex Stainless Steel
FPSO	Floating Production Storage and Offloading
GRE	Glass Reinforced Epoxy
NDT	Non-Destructive Testing
P&ID	Piping & Instrument Diagram
PMS	Piping Material Specifications
SDSS	Super Duplex Stainless Steel
SS	Stainless Steel
VMS	Valve Material Specification

1.5. CODES, STANDARDS AND FEDERAL REGULATIONS AND LAWS

Regulations, Codes and Standards are summarized as below:

Item No.	Code / Standard	Description
American Society of Mechanical Engineers (ASME)		
1.	ASME B31.3	Process Piping
2.	ASME B31.4	Pipeline Transportation Systems for Liquid and Slurries
3.	ASME B31.8	Gas Transmission and Distribution Piping Systems
4.	ASME B1.1	Unified Inch Screw Threads
5.	ASME B1.20.1	Pipe Threads, General Purpose (Inch)
6.	ASME B16.5	Pipe Flanges and Flanged Fittings NPS 12 through NPS 24 Metric / Inch Standard
7.	ASME B16.9	Factory-Made Wrought Steel Buttwelding Fittings
8.	ASME B16.10	Face-to-Face and End-to-End Dimensions of Valves
9.	ASME B16.11	Forged Fittings, Socket-Welding and Threaded
10.	ASME B16.20	Metallic Gaskets for Pipe Flanges Ring-Joint, Spiral Wound and Jacketed
11.	ASME B16.21	Non-metallic Flat Gaskets for Pipe Flanges
12.	ASME B16.25	Buttwelding Ends
13.	ASME B16.34	Valves- Flanged, Threaded, and Welding End
14.	ASME B16.36	Orifice Flanges
15.	ASME B16.47	Large Diameter Steel Flanges NPS 26 through NPS 60 Metric / Inch Standard
16.	ASME B16.48	Line Blanks
17.	ASME B16.49	Factory-Made, Wrought Steel, Buttwelding Induction Bends for Transportation and Distribution Systems
18.	ASME B18.2.1	Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
19.	ASME B 18.2.2	Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

Item No.	Code / Standard	Description
20.	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
21.	ASME B36.19M	Stainless Steel Pipe
22.	ASME SEC.VIII	Pressure vessel code
23.	ASME SEC.IX	Welding and brazing qualifications
American Petroleum Institute (API)		
1.	API SPEC 5L	Specification for Line Pipe
2.	API SPEC 6D	Specification for Pipeline Valves
3.	API SPEC 6A	Specification for Wellhead and Christmas Tree Equipment
4.	API SPEC 6FA	Specification for Fire Test for Valves
5.	API RP 14E	Recommended Practice for Design and Installation of Offshore Production Platform Piping Systems
6.	API RP 14G	Recommended Practice for Fire Prevention and Control on Fixed Open Type Offshore Production Platforms
7.	API RP 14J	Recommended Practice for Design and Hazards Analysis for Offshore Production Facilities
8.	API STD 594	Check Valves: Flanged, Lug, Wafer, and Butt-welding
9.	API STD 598	Valves Inspection and Testing
10.	API STD 600	Steel Gate Valves – Flanged and Butt-welding Ends, Bolted Bonnet
11.	API STD 602	Steel Gate, Globe, and Check Valves for Sizes NPS 4 (DN 100) and Smaller for Petroleum and Natural Gas Industries
12.	API STD 607	Fire Test for Quarter-turn Valves and Valves Equipped with Nonmetallic Seats
13.	API STD 609	Butterfly Valves: Double-flanged, Lug- and Wafer-type
British Standards (BS)		

Item No.	Code / Standard	Description
1.	BS 1873	BS 1873 Specification for Steel Globe and Globe Stop and Check Valves (Flanged and Butt-welding Ends) for Petroleum, Petrochemical, and Allied Industries
2.	BS 1868	Specification for Steel Check Valves (Flanged and Butt-welding Ends) for Petroleum, Petrochemical, and Allied Industries
3.	BS EN 12266-1	Industrial Valves- Testing of Metallic Valves Part 1: Pressure Test, Test Procedures and Acceptance Criteria – Mandatory Requirements
4.	BS EN 12266-2	BS EN 12266-2 Industrial Valves- Testing of Metallic Valves Part 2: Tests, Test Procedures and Acceptance Criteria Supplementary Requirement
5.	BS EN 10497	BS EN 10497 Testing of Valves- Fire Type- Testing Requirements
International Standardization Organization (ISO)		
1.	ISO 17292	Metal Ball Valves for Petroleum, Petrochemical, and Allied Industries
2.	ISO 15761	Steel Gate, Globe, and Check Valves for Sizes (DN 100) and Smaller for Petroleum and Natural Gas Industries
3.	ISO 10434	ISO 10434 Bolted Bonnet Steel Gate Valves for the Petroleum, Petrochemical and Allied Industries
4.	ISO 14313	Petroleum and Natural Gas Industries- Pipeline Transportation Systems – Pipeline Valves
Manufacturers Standardization Society (MSS)		
1.	MSS SP-25	MSS SP-25 Standard Marking System for Valves, Fittings, Flanges, and Unions
2.	MSS SP-44	Steel Pipeline Flanges
3.	MSS SP-75	Specification for High-Test, Wrought, Butt-Welding Fittings
4.	MSS SP-97	Integrally Reinforced Forged Branch Outlets Fittings- Socket Welding, Threaded, and Buttwelding Ends

Item No.	Code / Standard	Description
5.	MSS SP-58	Pipe hangers and supports – Materials, Design, Manufacture, Selection
National Association of Corrosion Engineers (NACE)		
1.	NACE MR 01-75 /ISO 15156-1/2/3	Material for use in H ₂ S containing environments in Oil and Gas production. Part-1: General principal for selection of cracking Resistant material. Part-2: Cracking resistant carbon and low carbon Steels and the use of cast irons.
2.	NACE TM-01-77	Laboratory-Testing of metals for resistance to sulphides cracking and stress corrosion cracking in H ₂ S environments.
3.	NACE TM-02-84	Evaluation of pipeline and pressure vessel steels for resistance to hydrogen induced cracking.
Norsok Standard		
1.	NORSOK L-001	Piping and valves
2.	NORSOK L-002	Piping system layout, design and structural analysis.
3.	NORSOK L-003	Piping details
4.	NORSOK L-004	Piping fabrication, installation, flushing and test specification
5.	NORSOK L-005	Compact Flanged Connections
American Society of Testing and Materials (ASTM)		
6.	A 105	Forgings, Carbon Steel, for Piping Components
7.	A 106	Seamless Killed Carbon Steel Pipe
8.	A 123	Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
9.	A 153	Zinc coating (hot dip) on iron & steel hardware

Item No.	Code / Standard	Description
10.	A 182	Forged or Rolled Alloy – Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High Temperature Service
11.	A 193	Specification for Alloy Steel and Stainless-Steel Bolting Materials for High-Temperature Service
12.	A 194	Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High Temperature Service
13.	A 216	Carbon Steel Castings Suitable for Fusion Welding for High-Temperature Service
14.	A 234	Specification for piping fittings of wrought carbon steel and alloy steel for moderate and elevated temperature
15.	A 262	Recommended practice for detecting susceptibility to intergranular corrosion attack in stainless steel
16.	A 307	Carbon Steel Bolts, Studs, and Threaded Rod 60000 PSI Tensile Strength
17.	A 312	Seamless and welded austenitic stainless steel
18.	A 320	Alloy-Steel and Stainless-Steel Bolting for Low-Temperature Service
19.	A 333	Seamless and Welded Steel pipe for Low-Temperature Services
20.	A 350	Carbon and Low-Alloy Steel Forgings, Requiring Notch Toughness Testing for Piping Components
21.	A 358	Electric-Fusion-Welded Austenitic Chromium-Nickel Stainless Steel Pipe for High-Temperature Service and General Applications
22.	A 370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
23.	A 403	Wrought Austenitic Stainless Steel Piping Fittings
24.	A 420	Piping Fittings of Wrought Carbon Steel and Alloy Steel for Low-Temperature Service
25.	A 453	High-Temperature Bolting, with Expansion Coefficients Comparable to Austenitic Stainless Steels
26.	A 488	Qualifications of Procedures and Personnel for the Welding of Steel Castings
27.	A 562	Pressure Vessel Plates, Carbon Steel, Manganese-Titanium for Glass or Diffused Metallic Coatings

Item No.	Code / Standard	Description
28.	A 578	Straight-Beam Ultrasonic Examination of Rolled Steel Plates for Special Applications
29.	A 671	Electric-Fusion-Welded Steel Pipe for Atmospheric and Lower Temperatures
30.	A 694	Carbon and Alloy Steel Forgings for Pipe Flanges, Fittings, Valves, and Parts for High-Pressure Transmission Service
31.	A 770	Through-Thickness Tension Testing of Steel Plates for Special Applications
32.	A 790	Seamless and Welded Ferritic/Austenitic Stainless-Steel Pipe
33.	A 815	Wrought Ferritic, Ferritic/Austenitic, and Martensitic Stainless Steel Piping Fittings
34.	A 860	Wrought High-Strength Ferritic Steel Butt-Welding Fittings
35.	A 928	Ferritic/Austenitic (Duplex) Stainless Steel Pipe Electric Fusion Welded with Addition of Filler Metal
36.	B 42	Standard specifications for seamless copper pipe
37.	B 124	Standard specification for copper and copper alloy forging rod, bar and shapes
38.	B 363	Seamless and Welded Unalloyed Titanium and Titanium Alloy Welding Fittings
39.	B 861	Titanium and Titanium Alloy Seamless Pipe
40.	D 1599	Standard Test Method for Resistance to Short-Time Hydraulic Pressure of Plastic Pipe, Tubing, and Fittings
41.	D 1784	Rigid Poly (Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds
42.	D 2992	Standard Practice for Obtaining Hydrostatic or Pressure Design Basis for Fiberglass (Glass Fiber Reinforced Thermosetting Resin)
43.	D 2996	Filament-wound Fiberglass (Glass fiber Reinforced Thermosetting Resin) Pipe
44.	D 4024	machine made Fiberglass (Glass fiber Reinforced Thermosetting Resin) Flanges
45.	D 5685	Fiberglass (Glass fiber Reinforced Thermosetting Resin) Pressure Pipe Fittings
46.	E 10	Standard Test Method for Brinell Hardness of Metallic Materials

Item No.	Code / Standard	Description
47.	E 18	Standard Test Methods for Rockwell Hardness of Metallic Materials ^{1, 2}
48.	E 45	Methods for Determining the Inclusion Content of Steel
49.	E 92	Standard Test Methods for Vickers Hardness and Knoop Hardness of Metallic Materials
50.	F 441	Chlorinated Poly Vinyl Chloride (CPVC) Plastic Pipe Schedules 40 and 80
51.	F 437	Threaded Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80
52.	F 439	Chlorinated Poly Vinyl Chloride (CPVC) Plastic Pipe Fittings Schedule 80
53.	G 36	Standard Practice for Evaluating Stress-Corrosion-Cracking Resistance of Metals and Alloys in a Boiling Magnesium Chloride Solution
54.	G 48	Standard Test Methods for Pitting and Crevice Corrosion Resistance of Stainless Steels and Related Alloys by Use of Ferric Chloride Solution
55.	ASTM A 269	Standard Specification for Seamless and Welded Austenitic Stainless-Steel Tubing for General Service
56.	ASTM F1387	Standard Specification for Performance of Piping and Tubing Mechanically Attached Fittings.

1.6. PROJECT SPECIFICATIONS

50501-SPE-80000-PP-ET-0002	Valve Material Specification
50501-SPE-80000-PP-RL-0001	Piping Design basis
TBA	General Specification for Paint and Protective Coating
50501-SPE-80000-PP-ET-0005	Piping Fabrication, Installation, Inspection and Testing
TBA	Preservation Procedure

1.7. CONFLICTING REQUIREMENT

The following is the order of precedence:

- Regulatory Requirements
- Classification Society
- International Codes & Standards
- COMPANY Project Technical Specifications, Drawings / COMPANY Standards
- CONTRACTOR Project Technical Specifications, Drawings / CONTRACTOR Project references

In case of any inconsistencies or conflicting requirements between the above, the CONTRACTOR shall indicate such to COMPANY and shall obtain COMPANY's written directive.

In the event of an inconsistency, conflict or discrepancy between any standards, CONTRACTOR's specifications, philosophies, datasheets, drawings, and regulatory requirements, the most stringent and safest requirement applicable to the project will prevail but to be taken up on a case-by-case basis, to the extent of inconsistency, conflict or discrepancy.

1.8. CONFLICTING REQUIREMENT

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1.9. DEVIATIONS

Any deviations, modifications and revisions to this specification shall only be made after first gaining approval in writing from the CONTRACTOR. Failure to observe this requirement may result in remedial work at Vendor's expense.

2. GENERAL

Materials shall be as per ASTM, BS or API specifications as referred in relevant PMS sheet and in compliance with the COMPANY Functional Specifications. Materials shall be new and unused, clean and free from rust, pits and obvious defects. Material older than two years from the date of manufacturing shall not be permitted.

2.1. MECHANICAL REQUIREMENTS

2.1.1. MINIMUM THICKNESS

The minimum thickness of material shall be based on requirements of pressure and other mechanical loading. Minimum wall thickness of carbon steel pipes, including corrosion allowance, shall be as following:

1. DN 40 (1 1/2") or less - schedule 80
2. DN 50 (2") through DN 150 (6") - schedule 80
3. DN 200 (8") through DN 600 (24") - schedule STD
4. Above DN 600 (24") - schedule STD

2.1.2. CORROSION ALLOWANCE

All Carbon Steel material shall incorporate a minimum corrosion allowance of 3 mm. Stainless steels, copper nickel, GRE material, and other corrosion resistant materials shall be used without extra-thickness for corrosion allowance.

2.1.3. MILL TOLERANCE

The mill tolerance on wall thickness shall be minus 12.5 %.

2.2. MATERIAL IDENTIFICATION AND CERTIFICATION DOCUMENTS

EN10204 Type 3.1/3.2, shall be provided as applicable. All piping components material certification shall be traceable to heat numbers. Certificates, including all mill certificates, material certificates, mechanical test certificates, welding qualification certificates, heat treatment certificates and hydrostatic test certificates duly certified by COMPANY approved third party inspection agency shall be available at final inspection and for counter signature by the certification authority and stored for a minimum of 5 years after acceptance of the piping by the CONTRACTOR.

2.3. SOUR SERVICE REQUIREMENTS

This shall be in accordance with NACE Standard MR-01-75/ISO 15156-1/2/3.

2.4. BOLTS AND NUTS

All bolting shall comply with applicable ASTM standards, ASME B16.5, and ASME B16.47 series A requirements.

Bolting shall conform to ASME B1.1 and the relevant ASTM standards. Nuts shall conform to ASME B18.2.2. All stud bolts shall be class 2A fit and nuts shall be class 2B fit. Nuts shall be semi-finished, heavy, hexagon.

Threads for all bolts and nuts up to 1 inch diameter shall be unified coarse UNC series and 8 UN series for diameter 1 1/8 inch & above.

Hydraulic Bolt tensioning shall be carried out on all bolts 1.1/8-inch diameter and above. Bolting for use with bolt tensioners shall have the normal length increased by one bolt diameter.

All stud bolts and nuts shall be XYLAR 2 + XYLAN 1070 coated with minimum combined thickness of 50µm.

2.5. PIPING MATERIAL SPECIFICATION NAMING PHILOSOPHY

Piping material specification naming is as per the below table:

PART 1 - RATING		PART 2 - MATERIAL		PART 3 – OPTIONAL IDENTIFIER	
CODE	DESC	CODE	DESC	CODE	DESC
A	150#	1	CS-3mm CA	N	NACE
B	300#	2	CS-6mm CA	L	Low Temp
D	600#	3	CS GALV-3mm CA	A	125 Barg Pressure
E	900#	4	CS GALV-1.5mm CA	B	200 Barg Pressure
F	1500#	5	CS GALV-6mm CA	C	325 Barg Pressure
G	2500#	6	CS Internally coated		
J	5000#	9	SS316		
K	10000#	10	SS316L		
T	Tubing	20	DSS		
		25	SDSS		
		30	90/10 CuNi		
		31	Copper		

PART 1 - RATING		PART 2 - MATERIAL		PART 3 – OPTIONAL IDENTIFIER	
CODE	DESC	CODE	DESC	CODE	DESC
		40	GRE		
		41	GRV (BONSTRAND Series 5000C)		
		42	CPVC		
		50	SS316L/SS316 Tubing		
		60	6 Mo Tubing		

APPENDIX 1: BRANCH TABLE

CHART -1 (CS, LTCS, SS, DSS, SDSS)

Branch Table as per API RP 14E

RUN PIPE	<= 1	T																
	1.5	T	T															
	2	W	T	T														
	3	W	W	T	T													
	4	W	W	T	T	T												
	6	W	W	W	T	T	T											
	8	W	W	W	W	T	T	T										
	10	W	W	W	W	W	T	T	T									
	12	W	W	W	W	W	T	T	T	T								
	14	W	W	W	W	W	W	T	T	T	T							
	16	W	W	W	W	W	W	T	T	T	T	T						
	18	W	W	W	W	W	W	W	T	T	T	T	T					
	20	W	W	W	W	W	W	W	T	T	T	T	T	T				
	22	W	W	W	W	W	W	W	W	T	T	T	T	T	T			
	24	W	W	W	W	W	W	W	W	T	T	T	T	T	T	T		
	30	W	W	W	W	W	W	W	W	W	T	T	T	T	T	T	T	
	30	W	W	W	W	W	W	W	W	W	T	T	T	T	T	T	T	T
		<= 1	1.5	2	3	4	6	8	10	12	14	16	18	20	22	24	30	32
BRANCH PIPE																		

LEGEND

W WELDOLET

T TEE

CHART -2 (CS GALV)

RUN PIPE	<= 1	T																
	1.5	T	T															
	2	H	T	T														
	3	H	H	H	T													
	4	H	H	H	T	T												
	6	H	H	H	T	T	T											
	8	H	H	H	W	T	T	T										
	10	H	H	H	W	W	T	T	T									
	12	H	H	H	W	W	T	T	T	T								
	14	H	H	H	W	W	W	T	T	T	T							
	16	H	H	H	W	W	W	T	T	T	T	T						
	18	H	H	H	W	W	W	W	T	T	T	T	T					
	20	H	H	H	W	W	W	W	T	T	T	T	T	T				
	24	H	H	H	W	W	W	W	W	T	T	T	T	T	T			
		<= 1	1.5	2	3	4	6	8	10	12	14	16	18	20	24			
BRANCH PIPE																		

LEGEND

H THREADOLET

W WELDOLET

T TEE

CHART -3 (CuNi)

RUN PIPE	<= 1	T																
	1.5	T	T															
	2	S	T	T														
	3	S	S	T	T													
	4	S	S	T	T	T												
	6	S	S	W	T	T	T											
	8	S	S	W	W	T	T	T										
	10	S	S	W	W	W	T	T	T									
	12	S	S	W	W	W	T	T	T	T								
	14	S	S	W	W	W	W	T	T	T	T							
	16	S	S	W	W	W	W	T	T	T	T	T						
	18	S	S	W	W	W	W	W	T	T	T	T	T					
	20	S	S	W	W	W	W	W	T	T	T	T	T	T				
	24	S	S	W	W	W	W	W	W	T	T	T	T	T	T			
	28	S	S	W	W	W	W	W	W	T	T	T	T	T	T	T		
	32	S	S	W	W	W	W	W	W	W	T	T	T	T	T	T	T	
	36	S	S	W	W	W	W	W	W	W	T	T	T	T	T	T	T	T
		<= 1	1.5	2	3	4	6	8	10	12	14	16	18	20	24	28	32	36
BRANCH PIPE																		

LEGEND

S SOCKOLET

W WELDOLET

T TEE BW

CHART -4 (GRE)

RUN PIPE	0.75	-													LEGEND				
	1	-	T													T EQUAL TEE			
	2	-	RT	T											RT REDUCING TEE				
	3	-	RT	RT	T									S REDUCING SADDLE					
	4	-	RT	RT	RT	T													
	6	-	S	S	RT	RT	T												
	8	-	S	S	RT	RT	RT	T											
	10	-	S	S	RT	RT	RT	RT	T										
	12	-	S	S	RT	RT	RT	RT	RT	T									
	14	-	S	S	RT	RT	RT	RT	RT	RT	T								
	16	-	S	S	RT	RT	RT	RT	RT	RT	RT	T							
	18	-	S	S	RT	RT	RT	RT	RT	RT	RT	RT	T						
	20	-	S	S	RT	RT	RT	RT	RT	RT	RT	RT	RT	T					
	24	-	S	S	RT	RT	RT	RT	RT	RT	RT	RT	RT	RT	T				
		<= 1	1.5	2	3	4	6	8	10	12	14	16	18	20	24				
																	BRANCH PIPE		

LEGEND

T EQUAL TEE

RT REDUCING TEE

S REDUCING SADDLE

APPENDIX 2: DETAIL PIPING MATERIAL SPECIFICATION